

Class-II tip-safe grouped-harvest and Gaussian-Carleson reduction

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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Purpose and scope

This note continues A13-CLASSII-NPC-CONE-MARTINGALE-INJECTION-BALANCE inside the existing A13 claim. It does not create a peer-level claim and does not close the controlled-shell one-use theorem. Its purpose is to separate three parts of the remaining problem which can now be proved from the one genuinely geometric part which cannot yet be proved.

First, the amount of raw current energy which one shell can causally remove is written as an exact nonlinear Tikhonov harvest. Its linearisation is the retained A11 resolvent, not a coefficient-blind source norm. Second, the complete conservative score, including coefficient differentiation, obeys a cutoff-uniform A12-compatible Carleson theorem. At the uncontrolled Gaussian base its two pieces improve further to dyadic rates N_j^{-1} and N_j^{-2} . Third, CAT(0) strong convexity is extended through the aggregate cone tip, and the centered $H^{-1-\kappa}$ quadratic remainder is paid globally with one Cameron–Martin/sextic allocation.

Two tempting shortcuts are also retired. Integrating the absolute full-score bound along a finite physical shell path produces superquadratic control powers and cannot meet arbitrary one-use allocations. More seriously, the often suggested full nonlinear remainder estimate $\|\delta_j - L_j h_j\| \lesssim N_j^{-3/2} \|h_j\|^2$ omits a coefficient-curvature term and is false. The exact remaining theorem is a tip-safe production Schur–Jacobi inequality which keeps that term inside the grouped current square.

Closed here. Exact nonlinear harvest and retained linear resolvent; full conservative-score Carleson bound; explicit uncontrolled-Gaussian N_j^{-1}/N_j^{-2} score tail; CAT(0) strong energy secant through the cone tip; floor-uniform physical endpoint-distance comparison; and a global strict-past $H^{-1-\kappa}$ quadratic-form bound with arbitrary control and sextic allocations.

Still open. The finite-cutoff production good/bad Schur–Jacobi inequality, including the coefficient-curvature remainder and causal Gaussian-rooted grouping; finiteness/identification of the cone-tip first variation; extension from regular one-shot controls to every declared finite-energy drift; one-use; Nelson; and all removal limits.

The fixed data are the hash-pinned $L = 16$ six-real production model, positive floor $e = 10^{-12}$, common real-even strict sharp cubes, and regular mutually orthogonal one-shot strict-past controls. Every statement below is cutoff-wise unless a uniform limit is explicitly stated.

1 The exact nonlinear harvest

Put every current component, spatial direction, heat dummy, and generator in the Hilbert direct sum with the production Q_{II} metric. At shell j let w_j be the past current and let $\delta_j(h)$ be the exact current increment reachable by the physical shell control h . Then the raw secant is simply

$$\mathcal{R}_j(h) = \frac{1}{2} \|w_j + \delta_j(h)\|^2 - \frac{1}{2} \|w_j\|^2. \quad (1.1)$$

For $\eta > 0$ define

$$\mathcal{H}_{\eta,j}(w) = \frac{1}{2}\|w\|^2 - \inf_h \left\{ \frac{1}{2}\|w + \delta_j(h)\|^2 + \eta\|h\|_{\mathcal{H}_j}^2 \right\}. \quad (1.2)$$

Thus, pathwise and without a Taylor expansion,

$$\mathcal{R}_j(h) + \eta\|h\|_{\mathcal{H}_j}^2 \geq -\mathcal{H}_{\eta,j}(w_j). \quad (1.3)$$

This is the energy which the declared control can actually harvest, rather than the total Gaussian energy injected at the scale. A sufficient global successor estimate is

$$\mathbb{E} \sum_{j \geq j_0} \mathcal{H}_{\eta,j}(w_j) \leq \zeta \mathbb{E} \|Z_J\|_6^6 + C_{\zeta,\eta,j_0} \quad (1.4)$$

uniformly in J and along every regular controlled trajectory.

If $\delta_j(h) = L_j h$ is linear and $s_j = L_j^* w_j$, completing the Hilbert square gives the exact retained resolvent

$$\mathcal{H}_{\eta,j}^{\text{lin}}(w) = \frac{1}{2} \left\langle s_j, (L_j^* L_j + 2\eta \mathbf{I})^{-1} s_j \right\rangle. \quad (1.5)$$

The executable solves the normal equation independently and checks (1.5). This is the same structural object as the A11 determinant resolvent. Replacing it by $\|s_j\|^2/(4\eta)$ deletes the retained operator and returns to the already failed coefficient-blind budget.

2 The full conservative-score Carleson theorem

Let

$$\mathcal{E}(u) = \frac{1}{2} \sum_i \int (D_i u)^T B(u) D_i u \, dx, \quad u_j = S_{j-1} \phi, \quad a_j = K_j h. \quad (2.1)$$

For a vector $q(u, Du) \in \mathbb{R}^6$ define it by duality,

$$\xi \cdot q(u, Du) = \sum_i DB(u)[\xi] : (D_i u \otimes D_i u). \quad (2.2)$$

Differentiating before any completion gives

$$D\mathcal{E}(u_j)[K_j h] = \langle \sigma_j(u_j), h \rangle, \quad \sigma_j = G_j^*[B(u_j)Du_j] + \frac{1}{2}K_j^*q(u_j, Du_j). \quad (2.3)$$

The factor 1/2 in the second term is compulsory.

Set

$$\beta_0 = \sup_{u \neq 0} \frac{\|B(u)\|_{\text{op}}}{|u|^2}, \quad \beta_1 = \sup_{u \neq 0, |\xi|=1} \frac{\|DB(u)[\xi]\|_{\text{op}}}{|u|}. \quad (2.4)$$

The fixed floor and the frame estimates

$$\|M_A(u)\|_F \leq \sqrt{20}|u|, \quad \|DM_A(u)[\xi]\|_F \leq \sqrt{68}|\xi| \quad (2.5)$$

give the safe production constants

$$\beta_0 = 4(a + 2b + c) = \frac{339}{2000P}, \quad \beta_1 \leq 6\sqrt{1360} \|Q_{\text{II}}\|_{\text{op}}, \quad P = 4 + 10^{-12}. \quad (2.6)$$

In particular $|q(u, Du)| \leq \beta_1 |u| |Du|^2$.

Let $A_0 = M_R^2/c_{\text{sym}}$ and retain the A12 sharp-cube constants M_6, Q_6 . The derivative source obeys the A12 estimate. For the coefficient source use

$$\|K_j^* F\|_2^2 \leq \frac{A_0}{(1 + \kappa_j^2)^2} \|F\|_2^2 \quad (2.7)$$

and

$$\sum_j \frac{|Du_j|^4}{(1 + \kappa_j^2)^2} \leq \left(\sum_j \frac{|Du_j|^2}{1 + \kappa_j^2} \right)^2. \quad (2.8)$$

Hölder and the A12 maximal/square-function theorem then prove

$$\boxed{\sum_j \|\sigma_j(u_j)\|_{\mathcal{H}_j}^2 \leq A_0 \left(\beta_0 M_6^2 Q_6 + \frac{\beta_1}{2} M_6 Q_6^2 \right)^2 \|\phi\|_6^6.} \quad (2.9)$$

This is a cutoff-uniform theorem for the complete linearised energy score, not only the old derivative source. It is qualitative: its fixed coefficient does not become an arbitrary ζ , so (2.9) alone does not close A13.

3 The uncontrolled Gaussian score is dyadically summable

There is extra decay at the uncontrolled Gaussian base which deterministic L^6 envelopes cannot see. Let $U_j = X_{<j} + r_{j:J}$ be the point value, including the spatially frozen terminal heat dummy, and let $Y_j = DX_{<j}$. Evenness of the common covariance makes U_j independent of Y_j at each spatial point. Define

$$S_* := \sup_{j,J} \text{Tr Cov}(U_j), \quad D_j := \text{Tr Cov}(Y_j). \quad (3.1)$$

Grouping lattice points by $\|n\|_\infty = m$ gives

$$S_* \leq \frac{6L^{-3}M_R^2}{c_{\text{sym}}} \left[1 + \sum_{m \geq 1} \frac{24m^2 + 2}{(1 + \alpha_L^2 m^2)^2} \right], \quad (3.2)$$

$$D_j \leq \frac{6L^{-3}M_R^2}{c_{\text{sym}}} \alpha_L^{-2} \left(24N + \frac{\pi^2}{3} \right), \quad N = N_{j-1}, \quad \alpha_L = \frac{2\pi}{L}. \quad (3.3)$$

For a centered Gaussian vector with covariance trace S , $\mathbb{E}|U|^4 \leq 3S^2$; similarly $\mathbb{E}|Y|^4 \leq 3D^2$. Jensen in the heat dummy, (2.7), and value-derivative independence yield

$$\mathbb{E}\|\ell_j\|^2 \leq \frac{3A_0\beta_0^2 L^3 S_*^2 D_j}{1 + \kappa_j^2} \leq \frac{C_\ell}{N}, \quad (3.4)$$

$$\mathbb{E}\|m_j\|^2 \leq \frac{3A_0\beta_1^2 L^3 S_* D_j^2}{4(1 + \kappa_j^2)^2} \leq \frac{C_m}{N^2}. \quad (3.5)$$

Here $\ell_j = G_j^*[B(U_j)Y_j]$ and $m_j = K_j^*q(U_j, Y_j)/2$. Safe explicit constants are obtained from

$$C_D = \frac{6L^{-3}M_R^2}{c_{\text{sym}}} \alpha_L^{-2} \left(24 + \frac{\pi^2}{3} \right), \quad (3.6)$$

$$C_\ell = 3A_0\beta_0^2 L^3 S_*^2 C_D \alpha_L^{-2}, \quad C_m = \frac{3}{4} A_0 \beta_1^2 L^3 S_* C_D^2 \alpha_L^{-4}. \quad (3.7)$$

Consequently

$$\mathbb{E} \sum_{j \geq j_0} \|\ell_j + m_j\|^2 \leq \frac{4C_\ell}{N_{j_0-1}} + \frac{8C_m}{3N_{j_0-1}^2}. \quad (3.8)$$

For every predictable infinitesimal feedback,

$$\mathbb{E} \sum_{j \geq j_0} \{\langle \ell_j + m_j, h_j \rangle + \eta \|h_j\|^2\} \geq -\frac{1}{4\eta} \mathbb{E} \sum_{j \geq j_0} \|\ell_j + m_j\|^2. \quad (3.9)$$

Thus no linear-response strategy around the base Gaussian can create a nonsummable ultraviolet loss. Equations (3.4)–(3.9) may not be transplanted to a controlled base: that would reuse the Gaussian independence after the control has destroyed it.

4 CAT(0) strong secants through the cone tip

Let $Y = C(B)$ be the complete aggregate production cone already identified in the preceding A13 result. For $U_0, U_1 \in W^{1,2}(\mathbb{T}_L^3; Y)$ let U_t be the pointwise constant-speed geodesic and put $d(x) = d_Y(U_0(x), U_1(x))$. The CAT(0) quadrilateral inequality applied at x and $x + he_i$, divided by h^2 and passed to the Korevaar–Schoen limit, proves

$$\boxed{\mathcal{E}(U_t) + \frac{t(1-t)}{2} \|\nabla d\|_2^2 \leq (1-t)\mathcal{E}(U_0) + t\mathcal{E}(U_1).} \quad (4.1)$$

This statement includes $R = 0$ and requires no smooth-stratum approximation. It also proves $d \in H^1$ and lower semicontinuity of the cone energy.

The function

$$t \mapsto \mathcal{E}(U_t) - \frac{t^2}{2} \|\nabla d\|_2^2 \quad (4.2)$$

is convex. Hence the right slope $\mathfrak{F}_+$ exists in the extended real line, and

$$\mathcal{E}(U_1) - \mathcal{E}(U_0) \geq \mathfrak{F}_+(U_0, U_1) + \frac{1}{2} \|\nabla d\|_2^2. \quad (4.3)$$

On the smooth $R > 0$ stratum this is the preceding Cartan–Jacobi formula. For a general tip-touching finite-energy pair, however, finiteness of $\mathfrak{F}_+$ and convergence of smooth first variations are not yet established. The safe tip theorem is therefore the whole finite secant (4.1), not a separately estimated logarithm.

There is also a global physical endpoint comparison. Write Z for the active doublet of X . The production pullback satisfies

$$|D\Phi_e(X)H|_g^2 = H^T B_e(X)H \leq \beta_0 |Z|^2 |H|^2. \quad (4.4)$$

The affine physical path $X_t = X + ta$ then gives, even if it crosses the tip,

$$\boxed{d_g(\Phi_e(X), \Phi_e(X+a)) \leq \sqrt{\beta_0} \left(|Z| + \frac{|aZ|}{2} \right) |a| \leq \sqrt{\beta_0} (|X| + |a|) |a|.} \quad (4.5)$$

The constant is floor-uniform and equals $\sqrt{339/[2000(4+10^{-12})]}$. In particular L^4 convergence of physical fields implies $L^2(Y)$ convergence of their cone images, so the energy lower semicontinuity in (4.1) applies directly at the tip. This controls endpoint distance; it does not control the spatial derivative of a cone logarithm.

5 A global centered-form one-use lemma

The balanced coefficient-jet result supplies every finite probability moment of the centered derivative-square model in $H^{-1-\kappa}$, with $0 < \kappa < 1$; the production audit uses $\kappa = 1/10$. Let $Q_{j-1} = S_{j-1}Q$ and let a_j be mutually orthogonal sharp-shell physical controls. Put

$$\Pi(a) = \sum_j S_{j-1}(a_j \otimes a_j), \quad X = \sum_j \|a_j\|_{H^2}^2, \quad A = \left(\sum_j |a_j|^2 \right)^{1/2}, \quad Y = \|A\|_6^6. \quad (5.1)$$

All Sobolev spaces here are inhomogeneous; this is load-bearing for zero modes. By self-adjointness of S_{j-1} , duality, and the sharp-cube maximal theorem at exponent $3/2$,

$$\|\Pi(a)\|_2 \leq C_L Y^{1/3}. \quad (5.2)$$

The $p = 3/2$ maximal estimate follows from the same de la Vallee Poussin plus cube-annular product-Marcinkiewicz proof used in A12. At the high endpoint, the three-dimensional H^2 algebra and L^2 boundedness of each sharp projection give

$$\|\Pi(a)\|_{H^2} \leq C_H X. \quad (5.3)$$

Interpolating (5.2)–(5.3), with $\vartheta = (1 + \kappa)/2$, yields

$$\boxed{\|\Pi(a)\|_{H^{1+\kappa}} \leq C X^\vartheta Y^{(1-\vartheta)/3}}. \quad (5.4)$$

Let $M = \|Q\|_{H^{-1-\kappa}}$. Duality and three-factor weighted Young now give, for every $\eta, \zeta > 0$,

$$\boxed{\left| \sum_j \langle Q_{j-1}, a_j \otimes a_j \rangle \right| \leq \eta X + \zeta Y + C_\kappa \eta^{-\frac{3(1+\kappa)}{2(1-\kappa)}} \zeta^{-1/2} M^{3/(1-\kappa)}}. \quad (5.5)$$

At $\kappa = 1/10$, the model moment is $10/3$ and the negative η exponent is $11/6$. Both are available from the balanced-jet theorem.

A1 ellipticity makes $X \lesssim \sum_j \|h_j\|_{\mathcal{H}_J}^2$. Moreover $a_j = P_j(Z_J - X_J)$, and the sharp-cube Littlewood–Paley theorem gives

$$\mathbb{E}Y \leq C\mathbb{E}\|Z_J\|_6^6 + C\mathbb{E}\|X_J\|_6^6 \leq C\mathbb{E}\|Z_J\|_6^6 + C. \quad (5.6)$$

Thus any centered quadratic remainder of the form on the left of (5.5) is globally one-use: it spends one control allocation, one terminal sextic allocation, and one integrable random constant. Applying Young shell by shell would create one constant per shell and is forbidden.

6 Why absolute score integration still fails

The full-score theorem does extend to current-shell inserted states

$$v_j = S_{j-1}\phi + t_j K_j h_j, \quad 0 \leq t_j \leq 1. \quad (6.1)$$

For $x = \|\phi\|_6$, $y = (\sum_j \|h_j\|^2)^{1/2}$ and $\delta = N_{j_0}^{-1}$, shell Bernstein and A1 smoothing give

$$\left\| \sup_j |v_j| \right\|_6 \leq M_6 x + C_a \delta y, \quad \left\| \left(\sum_j \frac{|Dv_j|^2}{1 + \kappa_j^2} \right)^{1/2} \right\|_6 \leq Q_6 x + C_d \delta y. \quad (6.2)$$

Substitution in (2.9) and integration over t control the absolute secant only by a constant times

$$x^3 y + \delta x^2 y^2 + \delta^2 x y^3 + \delta^3 y^4. \quad (6.3)$$

For any fixed $\delta > 0$, the last term cannot be bounded by $\eta y^2 + \zeta x^6 + C$ for all x, y . Even the first term has a fixed Young tradeoff between the two requested arbitrary allocations. Therefore direct absolute-score parolinearisation is a valid estimate but an invalid closure architecture. The grouped positive current/Jacobi square must be retained.

There is an even simpler obstruction to an initial-score-only comparison. In the flat scalar target $m(u) = u^2$, take

$$h = -u + \sqrt{u^2 + c}. \quad (6.4)$$

Then the finite target displacement is the constant c . Its distance gradient, energy secant, and Jacobi term all vanish, while the initial affine target displacement $2uh$ generally has nonzero gradient. Hence no universal finite-geodesic inequality may replace the endpoint displacement by the initial physical score with only $\mathfrak{J}$ as an error.

7 The exact scalar good/bad Schur prototype

The correct local structure is visible for

$$f(u, p) = u^2 p^2. \quad (7.1)$$

When $u + a \neq 0$, completing the endpoint derivative b gives exactly

$$\begin{aligned} & f(u + a, p + b) - f(u, p) - Df(u, p)[a, b] \\ &= (u + a)^2 \left[b + \frac{ap(2u + a)}{(u + a)^2} \right]^2 - \frac{a^2 p^2 u(3u + 2a)}{(u + a)^2}. \end{aligned} \quad (7.2)$$

If $|a| \leq \theta|u|$ with $0 \leq \theta < 1$, the sharp uniform bound is

$$f(u + a, p + b) - f(u, p) - Df(u, p)[a, b] \geq -\frac{3 - 2\theta}{(1 - \theta)^2} a^2 p^2, \quad (7.3)$$

uniformly in b . On the bad region $|u| \leq C|a|$, discard the tangent and use the whole secant:

$$f(u + a, p + b) - f(u, p) \geq -u^2 p^2 \geq -C^2 a^2 p^2. \quad (7.4)$$

Keeping the tangent globally is impossible. At $u = 1, a = -1$ its remainder is $p^2 - 2pb$, unbounded below as b varies. This is precisely why the cone geodesic and its tip behavior are load-bearing rather than cosmetic.

The production successor must prove a frame-valued analogue of (7.3)–(7.4), uniformly in the derivative displacement, while converting the $a_j^2 p_j^2$ remainder into the centered form of Section 5 plus the covariance trace. The latter is already harmless:

$$\langle \Gamma_{j-1}, a_j^2 \rangle \lesssim N_j \|a_j\|_2^2 \lesssim N_j^{-3} \|h_j\|_{\mathcal{H}_j}^2. \quad (7.5)$$

8 A production-scaled gauge-null stress test

The initial physical score can vanish even when a finite secant is negative. On a normalized circle take one real active component

$$u = C + A \cos(qx), \quad h = if, \quad f = B(\cos(nx) - \cos(mx)), \quad q = m - n. \quad (8.1)$$

Assume $m \neq 2n, 3n$, so no unintended resonance meets q or $2q$. The affine moment-map and density variations vanish because h is a phase tangent, but the affine endpoint changes the density by f^2 . With $I = \langle (uu_x)^2 \rangle$, Fourier orthogonality gives

$$\Delta I = -\frac{1}{2} AC q^2 B^2 + \frac{3}{8} (n^2 + m^2) B^4. \quad (8.2)$$

The production J -sector secant, including its full positive square, is

$$\Delta E_J = -aACq^2B^2 + \frac{3a}{4}(n^2 + m^2)B^4. \quad (8.3)$$

If the CM symbol lower bound is $\kappa_4|k|^4$, optimization in B^2 yields

$$\inf_B \{\Delta E_J + \eta \|h\|_{\mathcal{H}}^2\} = -\frac{L_+^2}{3a(n^2 + m^2)}, \quad (8.4)$$

$$L = aACq^2 - \frac{\eta\kappa_4}{2}(n^4 + m^4). \quad (8.5)$$

For $n = N + 1, m = 2N, q = N - 1, N \geq 2$, negativity requires $AC \gtrsim \eta N^2$. A production Gaussian q -mode has standard deviation $O(N^{-2})$. With bounded C this is an N^4 -standard-deviation event; if C is also Gaussian, the optimized joint tail is still $\exp(-cN^4)$. Alternatively, finite moments give after N^3 multiplicity the bound N^{9-4p} for any $p > 9/4$. The mechanism is therefore summable. It refutes initial-score-only reasoning but is not a production counterexample.

9 The corrected nonlinear remainder and exact successor

For

$$\delta = M(z + a)^T(y + b) - M(z)^T y \quad (9.1)$$

the true linearisation is

$$Lh = DM(z)[a]^T y + M(z)^T b. \quad (9.2)$$

The exact nonlinear remainder has two terms:

$$\boxed{\delta - Lh = [M(z + a) - M(z) - DM(z)a]^T y + [M(z + a) - M(z)]^T b.} \quad (9.3)$$

Only the second term obeys the attractive shell bound

$$\|[M(z + a) - M(z)]^T b\| \lesssim N_j^{-3/2} \|h_j\|^2. \quad (9.4)$$

The first is the frame-curvature/Cartan term. A naive fixed-floor Taylor bound at Gaussian derivative size is only of order $N_j^{-1/2} \|h_j\|^2$, and cone coordinates become singular near the tip. Thus any proof which assigns (9.4) to the whole remainder is invalid. The term must remain inside the grouped current/Jacobi square and be reduced to the centered form (5.5) by a good/bad Schur argument.

The remaining finite-cutoff lemma is now precise. It must combine:

1. a smooth good-region tangent where the active radius dominates the control displacement;
2. a tip-safe bad-region whole secant, with no forbidden b -linear tangent;
3. a Schur complement uniform in b_j which leaves only the centered Q_{j-1} form and the summable trace (7.5); and
4. a causal grouping in which pure-control current creation telescopes, while only Gaussian-rooted pieces use the N_j^{-1}/N_j^{-2} tail.

Once this grouped production Schur–Jacobi lemma is proved, Sections 3–5 provide the required arbitrary η, ζ allocations for regular one-shot controls. Localization and lower semicontinuity must then extend the result to the declared finite-energy class before one-use or Nelson can be invoked.

10 Devil’s-advocate review

1. **“The full-score Carleson theorem closes A13.” UPHELD as false.** Its coefficient is fixed, and finite path integration creates (6.3). It proves the complete linear score, not the required grouped finite secant.
2. **“The Gaussian score tail applies after adapted control.” UPHELD as false.** Equations (3.4)–(3.9) use same-point Gaussian value–derivative independence. The production successor must first isolate Gaussian-rooted terms without differentiating the control.
3. **“The cone tip was already harmless because the floor is positive.” DISMISSED in the stated form.** The density floor does not prevent the active doublet radius from vanishing. The whole-secants inequality (4.1), not the floor, supplies the tip-safe result. A finite separate first variation at the tip remains open.
4. **“The centered form can be paid shell by shell.” UPHELD as false.** That creates one Young constant per shell. Equations (5.1)–(5.6) sum first and pay the random model norm once.
5. **“The full nonlinear remainder has shell power $-3/2$.” UPHELD as false.** Equation (9.3) displays the omitted coefficient- curvature term. Only the product remainder has that power.
6. **“The gauge-null beat is a production counterexample.” DISMISSED.** Its CM threshold is $AC \gtrsim \eta N^2$, and every base-Gaussian tail estimate is summable after shell multiplicity. A control-created carrier also pays its creation energy in the global telescope.
7. **“CAT(0) strong convexity alone proves one-use.” UPHELD as false.** The previous flat reset no-go remains valid. Production Fourier/CM decay and the grouped Schur–Jacobi comparison are essential.

11 Reproduction and references

The fail-closed command is

```
python codes/foundations/  
a13_classii_tip_safe_grouped_harvest_carleson_reduction_verify.py
```

The primary route derives production constants, explicit Gaussian tail constants, the retained resolvent, scalar Schur completion, gauge beat, and Young exponents. The non-importing route reconstructs rational production coefficients, uses different finite-dimensional coefficients, exact rational Schur grids, an independent Fourier grid, Wick identities, and weighted AM–GM. The integrated verifier pins both scripts, their JSON artifacts, this source, the rendered PDF, all claim/ledger boundaries, and the negative-result entry.

The CAT(0) Sobolev-energy passage uses the metric energy and quadrilateral framework of N. Korevaar and R. Schoen, “Sobolev spaces and harmonic maps for metric space targets,” *Communications in Analysis and Geometry* 1 (1993), 566–601, and the lower-semicontinuity/differential formulation of N. Gigli and A. Tyulenev, “Korevaar–Schoen’s energy on strongly rectifiable spaces,” *Calculus of Variations* 60 (2021), article 235, <https://doi.org/10.1007/s00526-021-02028-z>.

Result footer

Result ID: A13-CLASSII-TIP-SAFE-GROUPED-HARVEST-CARLESON-REDUCTION
Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION
Tier: T4 before / T4 after (no promotion)
Closed: nonlinear Tikhonov harvest; retained linear resolvent;
full conservative-score Carleson theorem; uncontrolled
Gaussian N^{-1}/N^{-2} score tail; CAT(0) cone-tip strong
secant; physical endpoint distance; global centered-form
one-use bound; scalar good/bad Schur prototype
Open: production tip-safe grouped Schur--Jacobi lemma; finite
cone-tip first variation; finite-energy extension
Umbrella: A13-CLASSII-CONTROLLED-SHELL-ENERGY-ONE-USE remains open
Scope: fixed floor; finite cutoff; strict one-shot sharp shells;
regular controls; inhomogeneous H2; $0 < \kappa < 1$
No claim: global injection balance; one-use; Nelson; measure;
floor/regulator/infinite-volume limits; T5--T7

Final boundary. The stochastic, harmonic-analytic, centered-form, and cone-tip pieces of the successor are now individually controlled. The unresolved object is no longer a generic geodesic-log Carleson estimate: it is the finite-cutoff production good/bad Schur--Jacobi inequality which must keep the first term of (9.3) grouped with the positive current square and telescope pure-control current creation before applying the Gaussian tail.